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HT Board Booster

Class X Science

Chemistry

CBSE Sample Paper Question Bank

Chapter wise Practice

Chapters 1 - 2

Session 2026-27

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Class X Chemistry
CBSE SQP 2019-2025
Ch.- 1 Chemical reactions and equations

SECTION - A

Q.1. Which of the following reactions is an endothermic reaction? **(CBSE SQP 2019-20)**

i) Burning of coal.
ii) Decomposition of vegetable matter into compost.
iii) Process of respiration.
iv) Decomposition of calcium carbonate to form quick lime and carbon dioxide

Q.2. List any two observations when Ferrous Sulphate is heated in a dry test tube?

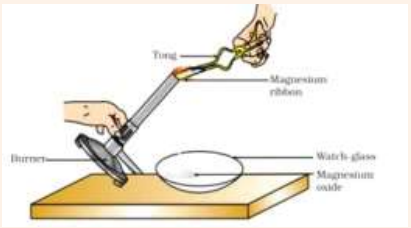
OR

Identify the products formed when 1 mL of dil. Hydrochloric acid is added to 1g of Sodium metal? **(CBSE SQP 2020-21)**

Q.3. In the redox reaction $\mathbf{MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2}$ **(CBSE SQP 2022-23)**

(a) MnO_2 is reduced to $MnCl_2$ & HCl is oxidized to H_2O
(b) MnO_2 is reduced to $MnCl_2$ & HCl is oxidized to Cl_2
(c) MnO_2 is oxidized to $MnCl_2$ & HCl is reduced to Cl_2
(d) MnO_2 is oxidized to $MnCl_2$ & HCl is reduced to H_2O

Q.4. Which of the following is the correct observation of the reaction shown in the following set up? **(CBSE SQP 2022-23)**

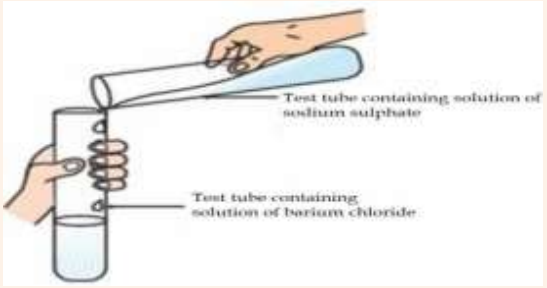


(a) Brown powder of Magnesium oxide is formed.
(b) Colourless gas which turns lime water milky is evolved.
(c) Magnesium ribbon burns with brilliant white light.
(d) Reddish brown gas with a smell of burning Sulphur has evolved.

Q.5. **Assertion:** Silver bromide decomposition is used in black and white photography.
Reason: Light provides energy for this exothermic reaction. **(CBSE SQP 2022-23)**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, and R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true

Q.6. Identify the product which represents the solid state in the above reaction. **(CBSE SQP 2023-24)**



(a) Barium chloride
(b) Barium sulphate
(c) Sodium chloride
(d) Sodium sulphate

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Q.7. The colour of the solution observed after 30 minutes of placing zinc metal to copper sulphate solution is **(CBSE SQP 2023-24)**

a) Blue b) Colourless c) Dirty Green d) Reddish Brown

Q.8. **Assertion:** Rusting of Iron is endothermic in nature. **(CBSE SQP 2023-24)**

Reason: As the reaction is slow, the release of heat is barely evident.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, and R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true

Q.9. Identify 'p', 'q' and 'r' in the following balanced reaction **(CBSE SQP 2024-25)**

$$p\ Pb(NO_3)_2(s) \xrightarrow{\text{Heat}} q\ PbO(s) + r\ NO_2(g) + O_2(g)$$

a) 2,2,4 b) 2,4,2 c) 2,4,4 d) 4,2,2

Q.10. When 50g of lead powder is added to 300 ml of blue copper sulphate solution, after a few hours, the solution becomes colourless. This is an example of **(CBSE SQP 2024-25)**

a) Combination reaction b) Decomposition reaction

c) Displacement reaction d) Double displacement reaction

Q.11. Which of the following is an endothermic reaction? **(CBSE SQP 2024-25)**

a) Burning of candle. b) Cooking of food.

c) Decomposition of Vegetable matter. d) Reaction of Sodium with air

Q.12. Which of the following equations represent redox reactions and what are the values for 'p' and 'q' in these equations? **(CBSE SQP 2025-26)**

Equation 1: $Fe_2O_3(s) + 2Al(s) \rightarrow Al_2O_3(s) + p\ Fe(l) + heat$

Equation 2: $2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + q\ H_2O(g)$

A. Only equation 1 is a redox reaction, p =1 and q =3

B. Both equations 1 and 2 are redox reactions, p = 2 and q =4

C. Only equation 2 is a redox reaction, p = 2 and q = 10

D. Both equations 1 and 2 are redox reactions, p = 2 and q =10

Q.13. In the reaction of aqueous solution of barium chloride with aqueous solution of sodium sulphate, the aqueous solution formed will be: **(CBSE SQP 2025-26)**

a) $BaCl_2$ b) $BaSO_4$ c) Na_2SO_4 d) NaCl

Section – B

Q.14. A clear solution of slaked lime is made by dissolving $Ca(OH)_2$ in an excess of water. This solution is left exposed to air. The solution slowly goes milky as a faint white precipitate forms. Explain why a faint white precipitate forms, support your response with the help of a chemical equation.

OR

Keerti added dilute Hydrochloric acid to four metals and recorded her observations as shown in the table given below: **(CBSE SQP 2022-23)**

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Metal	Gas evolved
Copper	Yes
Iron	Yes
Magnesium	No
Zinc	Yes

Select the correct observation(s) and give chemical equation(s) of the reaction involved.

Q.15. Dil. HCl is added to Zn granules. How will you prove that chemical change has taken place here? Support your response with two arguments. **(CBSE SQP 2023-24)**

Q.16. Identify the type of each of the following reactions stating the reason for your answers.

(a) $Fe_2O_3 + 2AI \rightarrow Al_2O_3 + 2Fe + heat$ **(CBSE SQP 2024-25)**

(b) $Pb(NO_3)_2 + 2KI \rightarrow PbI_2(\downarrow) + 2KNO_3$

Section – C

Q.17. (i) Write two observations when lead nitrate is heated in a test tube. **(CBSE SQP 2019-20)**

(ii) Name the type of reaction.

(iii) Write a balanced chemical equation to represent the above reaction.

Q.18. (a) Which of the following reactions is/ are an endothermic reaction(s) where decomposition also happens? **(CBSE SQP 2020-21)**

- Respiration
- Decomposition of organic matter
- Heating of lead nitrate
- Electrolysis of acidified water

(b) Silver chloride when kept in the open turns grey. Illustrate this with a balanced chemical equation.

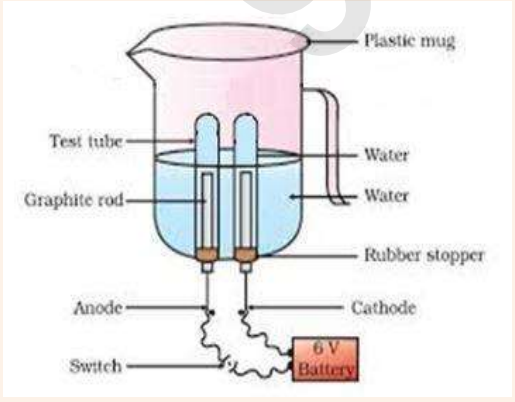
Q.19. Identify the types of reaction mentioned above in (i) and (ii). **(CBSE SQP 2022-23)**

i) $A + BC \rightarrow AC + B$

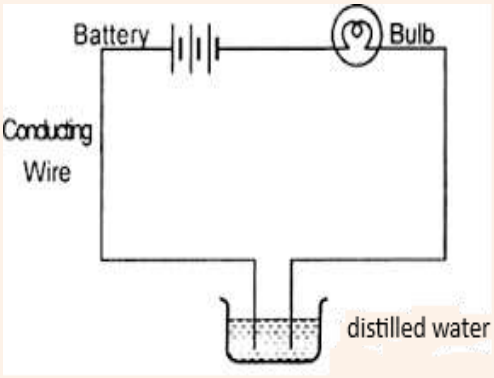
ii) $AB + CD \rightarrow AC + BD$

Give one example for each type in the form of a balanced chemical equation.

Q.20. Amrita electrolysed distilled water using the set-up shown in figure 1. She was expecting two gases to be evolved at the anode and cathode respectively



(fig.1)



(fig.2.)

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Suddenly, she realised that the bulb in the circuit did not glow when she used distilled water (figure 2). After this realization, she added a substance to the distilled water for electrolysis to take place. **(CBSE SQP 2025-26)**

Answer the following questions based on the information given above:

a) Which gas was she expecting to be formed at the anode and which one at the cathode respectively?

b) Why did the bulb not glow when Amrita passed electricity through distilled water?

c) Which substance was added by Amrita to distilled water to get the expected result?

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Class X Chemistry
CBSE SQP 2019-2025
Ch. - Acid Base and Salt

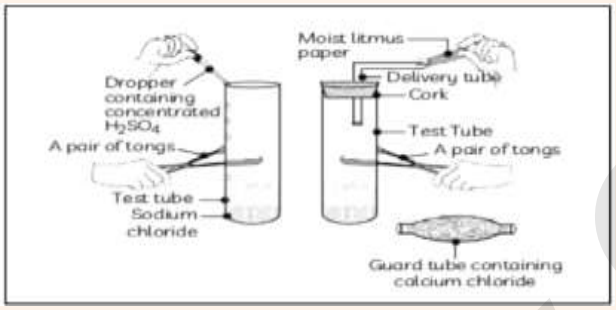
Section - A

Q.1. Identify the basic salt from the following salts: **(CBSE SQP 2019-20)**

(a) Na_2CO_3 (b) NH_4Cl (c) $NaNO_3$ (d) KCl

Q.2. Write the chemical name and chemical formula of the salt used to remove permanent hardness of water. **(CBSE SQP 2020-21)**

Q.3. The Change in colour of the moist litmus paper in the given set up is due to: **(CBSE SQP 2022-23)**



(i) presence of acid
(ii) presence of base
(iii) presence of $H^+(aq)$ in the solution
(iv) presence of Litmus which acts as an indicator

(a) i and ii (b) Only ii (c) Only iii (d) Only iv

Q.4. With the reference to four gases CO_2, CO, Cl_2 and O_2, which one of the options in the table is correct? **(CBSE SQP 2022-23)**

Option	Acidic Oxide	Used in Treatment of water	Product of respiration	Product of incomplete combustion
(a)	CO	Cl_2	O_2	CO
(b)	CO_2	Cl_2	CO_2	CO
(c)	CO_2	O_2	O_2	CO_2
(d)	CO	O_2	CO_2	CO_2

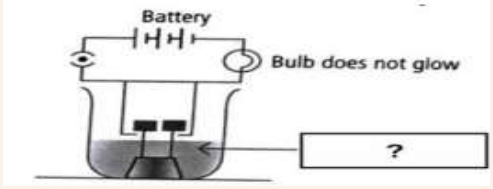
Q.5. Anita added a drop each of diluted acetic acid and diluted hydrochloric acid on pH paper and compared the colors. Which of the following is the correct conclusion? **(CBSE SQP 2022-23)**

(a) pH of acetic acid is more than that of hydrochloric acid
(b) pH of acetic acid is less than that of hydrochloric acid.
(c) Acetic acid dissociates completely in aqueous solution
(d) Acetic acid is a strong acid

Q.6. Mild non-corrosive basic salt is _____. **(CBSE SQP 2023-24)**

(a) Ca $(OH)_2$ (b) NaCl (c) NaOH (d) $NaHCO_3$

Q.7. The solution in the given figure is likely to be _____. **(CBSE SQP 2024-25)**



(a) HNO_3
(b) C_2H_5OH
(c) H_2SO_4
(d) CO_2 in water

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Q.8. An aqueous solution 'A' turns the phenolphthalein solution pink. On addition of an aqueous solution 'B' to 'A', the pink colour disappears. Which of the following statement is true for the solutions 'A' and 'B'?? **(CBSE SQP 2024-25)**

(a) A is strongly basic and B is a weak base
(b) A is strongly acidic and B is a weak acid
(c) A has a pH greater than 7 and B has a pH less than 7
(d) A has a pH less than 7 and B has a pH greater than 7.

Q.9. Methyl orange is added to dilute hydrochloric acid and to aqueous sodium hydroxide. What is the colour of the methyl orange in each solution? **(CBSE SQP 2025-26)**

Sample	**Colour in dilute hydrochloric acid**	**Colour in aqueous sodium hydroxide**
A	Orange	Red
B	Red	Yellow
C	Red	Orange
D	Yellow	Red

Q.10. Which of the following substances when dissolved in equal volume of water, will have the highest pH value? **(CBSE SQP 2025-26)**

(a) Sulphuric acid (b) Acetic acid
(c) Magnesium hydroxide (d) Sodium hydroxide

Section-C (3 Marker Each)

Q.11. A compound 'X' of sodium is used as an antacid and it decomposes on strong heating.

(i) Name the compound 'X' and give its chemical formula.
(ii) Write a balanced chemical equation to represent the decomposition of 'X'.
(iii) Give one use of compound 'X' besides an antacid.

OR

You are provided with 90 mL of distilled water and 10 mL of concentrated sulphuric acid to prepare dilute sulphuric acid. **(CBSE SQP 2019-20)**

(i) What is the correct way of preparing dilute sulphuric acid? Give reason.
(ii) How will the concentration of H_3O + ions change on dilution?

Q.12. Match the following pH values *1, 7, 10, 13* to the solutions given below:

- Milk of magnesia **(CBSE SQP 2020-21)**
- Gastric juices
- Brine
- Aqueous Sodium hydroxide.

Amit and Rita decided to bake a cake and added baking soda to the cake batter. Explain with a balanced reaction, the role of the baking soda. Mention any other use of baking soda.

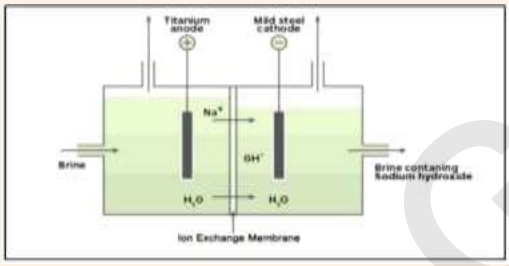
OR

(i) Four samples A, B, C and D change the colour of pH paper or solution to Green, Reddish-pink, Blue and Orange. Their pH was recorded as 7, 2, 10.5 & 6 respectively. Which of the samples has the highest amount of Hydrogen ion concentration? Arrange the four samples in the decreasing order of their pH.

(ii) Rahul found that the Plaster of Paris, which he stored in a container, has become very hard and lost its binding nature. What is the reason for this? Also, write a chemical equation to represent the reaction taking place.

(iii) Give any one use of Plaster of Paris other than for plastering or smoothening of walls

Q.13. (a) Identify the gasses evolved at the anode and cathode in the above experimental set up. **(CBSE SQP 2022-23)**



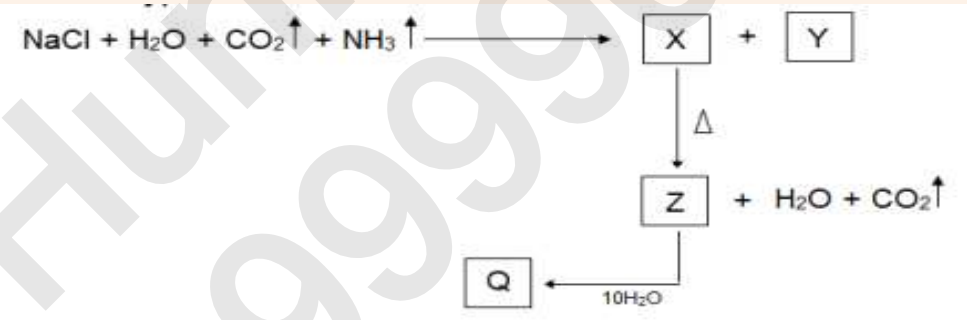
(b) Name the process that occurs. Why is it called so?

(c) Illustrate the reaction of the process with the help of a chemical equation.

Attempt either option A or B. **(CBSE SQP 2024-25)**

Q.14. **(a)**

(i) In the given series of reactions, name the compounds X and Z.



(ii) Which type of reaction is X to Z?

(iii) You are given 3 unknown solutions A, B, and C with pH values of 6, 8 and 9.5 respectively. In which solution will the maximum number of hydronium ions be present? Arrange the given samples in the increasing order of H^+ ion concentration.

OR

(b) Comment on the following statements:

(i) Bee sting is treated with baking soda paste whereas wasp sting is treated with dilute vinegar.

(ii) Farmers treat soil with quicklime when tilling.

(iii) Ancient sculptures and marble structures are conserved by treating them with certain chemicals."

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Section-D

Q.15. Sara took 2 mL of dilute NaOH solution in a test tube and added two drops of phenolphthalein solution to it. The solution turned pink in colour. She added dilute H_2SO_4 to the above solution drop by drop until the solution in the test tube became colourless. 40 drops of dilute H_2SO_4 were used for the change in colour from pink to colourless. When Sara added a drop of NaOH to the solution, the colour changed to back to pink again. Sara now tried the activity with different volumes of NaOH and recorded her observation in the table given below: **(CBSE SQP 2025-26)**

S.No	Volume of dil. NaOH taken (mL)	Drops of dil.H_2SO_4 used
1	2	20
2	3	30
3	4	40

Answer the following questions based on the above information:

A. If Sara used concentrated H_2SO_4 in place of dilute H_2SO_4, how many drops will be required for the change in colour to be observed?

(a) 40 (b) < 40 (c) >40

Justify your answer.

B. Sara measured 20 drops of dil. H_2SO_4 and found its volume to be 1 mL. If Sara observed a change in colour of NaOH solution by using 3 mL of H_2SO_4, how many mL of NaOH did she add to the test tube initially?

OR

Sara takes 10 drops of dilute H_2SO_4 in the test tube and adds two drops of phenolphthalein solution to it. Then she adds NaOH drop wise. Sara observes a change in colour after adding 20 drops of NaOH. What change in colour would she observe and why?

C. Write a balanced chemical equation for the reaction taking place in the above experiment. Which of the following is true and why? The reaction is a

(a) neutralisation and double displacement reaction
(b) neutralisation and precipitation reaction
(c) precipitation and double displacement reaction
(d) neutralisation, double displacement as well as precipitation reaction.

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